

Productivity evaluation indicators based on Lean and their application to compare agile and waterfall projects

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Abstract— Despite the increasing importance of responding quickly to markets and customers, many organizations that are contracted to develop software are still unable to move from waterfall to agile development. Major reasons include not only the incompatibility of current labor laws with agile development but also the inability to determine the productivity (cost, duration) of agile development. This paper proposes indicators to evaluate and compare the productivity of projects by focusing on "value building" in Lean process evaluation. To promote agile development, our laboratories have been running agile development trials using delegated agreements (similar to time and material contracts) since 2018. Using the proposed indicators, we compared the statistics of 20 agile trial projects and more than 200 waterfall development projects. Results revealed that only 5% improvement in feature-used rate in agile development could lead more efficiency in terms of delivery time and cost than that of waterfall development.

Keywords— Agile development, Productivity, Lean, Value

I. INTRODUCTION

While the importance of responding quickly to markets and customer is increasing more and more worldwide [1], organizations that have traditionally done waterfall development to fulfil contracts have often been unable to adopt agile development. Our organization is no exception and has long been hesitant to adopt a strategic approach. There have been two implementation challenges:

- Since current laws (especially in Japan) are based on the assumption waterfall development being used to fulfil software development contracts, the organizations ordering software development need to improve their operations and systems to correspond to current labor regulations, which in Japan include the Worker Dispatching Act and the Subcontracting Act.
- The lack of a defined productivity (which in this paper means development period and cost) evaluation methodology for agile development has prevented comparisons with other projects and historical data such as that of waterfall development. As a result, we have not been able to evaluate product quality or project capability objectively.

To overcome the first problem, contract forms [2] [3] for observing present laws and examinations of in-house operation system design [4] are being made. In this case, delegated agreements tend to be used [4], which has a high affinity with agile development. However, delegated agreement contracts do not require completion, which increases the risk to the ordering organization. For ordering organizations to accept agile development's shortcomings and adopt and promote it, agile development must be more productive than traditional contracting and development practices such as waterfall.

There are two main factors that potentially make agile development less productive than waterfall development:

- Person-month payment remains high, because the members of the organization entrusted with software development require the skills of a full stack engineer.
- Frequent software releases increase validation activities and their cost.

Given these potentially negative factors, the challenge is to prove the advantages of agile development in terms of cost and development period.

In this paper, we focus on "value building" in Lean and define three parameters that contribute to productivity that can be commonly used regardless of development methodologies such as waterfall and agile. We also propose indicators to evaluate productivity by using data of both the cost and development period of multiple past projects. In addition, all our laboratories are promoting agile development and have been conducting trials (agile trials) since August 2018 [4]. As a result of applying the proposed method to agile trial cases, the effectiveness of the indicators was confirmed. Using data from agile trials and past waterfall development projects, we have also found out the feature-used rate necessary to make agile development more productive than waterfall development. The remainder of this paper is as follows. Chapter II discusses the related works. Chapter III presents the proposed indicators. Chapter IV compares the results of applying the indicators to agile trial and the waterfall development projects. Finally, Chapter V presents the summary and future work.

II. RELATED WORKS

Productivity was traditionally evaluated by cost per production quantity such as line number and function point [5]: the higher the production volume per cost, the higher the productivity. However, with the spread of object-oriented programming languages, high productivity and high software cohesiveness [5] have come to be synonymous, and production quantity has become unsuitable as an indicator of productivity.

On the other hand, many metrics for developers have been proposed [6 -12] that mainly measure velocity, such as WIP (work in progress). However, these metrics do not directly measure productivity related to costs and periods and are closed indicators for projects, making it difficult to compare them with other projects including those in the past.

To overcome the obstacles to adopting agile development, we have developed an agile development process [13], examined the QCD (quality, cost, and delivery) for agile development based on the Lean concept [14] [15], and proposed a method to predict the development time for agile development with the aim of supporting portfolio management [16]. However, it does not directly refer to the productivity evaluation.

III. PROPOSED PRODUCTIVITY INDICATORS

The requirements for the proposed indicators are summarized below:

- Can assess productivity in terms of cost and duration
- Can be compared and evaluated against multiple cases

In this chapter, we propose productivity indicators that meet the above requirements and utilize the concept of Lean process evaluation technique.

A. Introduction of Lean concept

The basic idea of Lean is to thoroughly eliminate wasteful work unrelated to "value". Therefore, work related to value building is defined as VA (Value Added), work unrelated to value building is defined as NVA (Not Value Added), and work unrelated to value building but necessary is defined as NNVA (Necessary but Not Value Added) [14] [15]. Since work is the basis, work effort (time) will be used as the basic unit for productivity measurement.

We propose seven "work categories" in Table 1 corresponding to VA, NVA, and NNVA: consideration, production, verification, management and communication, environment preparation, documentation, and defect fixing. Each work category is a modified version of categories in the literature [16].

TABLE I. MODIFIED WORK CATEGORIES

WORK CATEGORIES	Examples	Candidate Lean categories
Consideration (VC: Value construction)	User story building, design, specifications, for functional/non-functional requirements, including related document review, etc.	VA
Production (VC: Value construction)	Detail design, coding, code review, unit testing, deployment, etc.	VA
Verification (VF)	Functional/non-functional testing, regression testing, field testing for whole software, etc.	NNVA
Management and communication (MC)	Overall communication including project management, authorization, meetings, etc.	NNVA
Environment preparation (EP)	Development/testing/staging/delivering environment preparation, procurement, etc.	NNVA
Documentation (DC)	Manual/instruction document, planning document, etc.	NNVA
Defect fixing (DF)	Failure detection and analysis, failure fixing, relevant explanation to stakeholders, after delivery and getting feedback.	NVA

B. Basic concept of productivity

Figure 1 shows the concept of the proposed productivity indicator. We suppose functions are developed with an actual

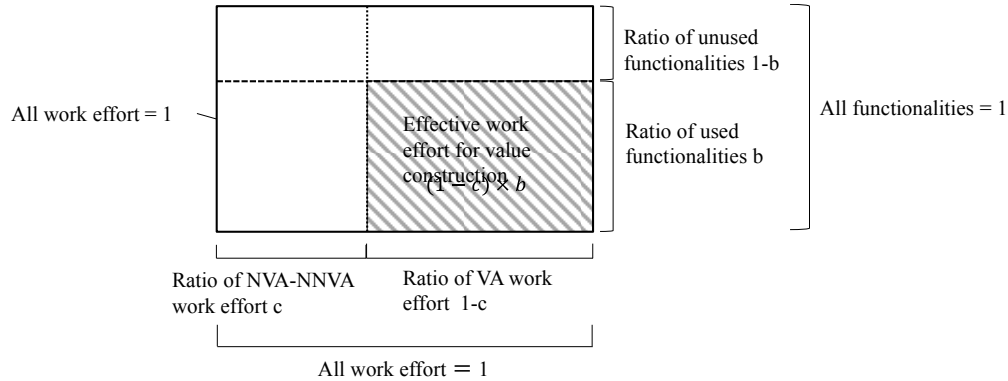


Fig.1 Concept of productivity evaluation

effort of 1. Here, on the basis of the Lean concept, we focus on two kinds of effort that do not create value.

- Work effort unrelated to value building in development
- Work effort related to building features that have not been used since the release

Assuming that the actual effort is 1, the effort that contributes to the value construction and that is actually used for the function is defined as “effective work effort” from the above two viewpoints. The effective work effort corresponds to the shaded portion in Figure 1. This means that the necessary effort equivalent to the inverse of effective effort is required to achieve effective value 1. In addition, the cost to achieve effective value 1 can be calculated by multiplying person-month payments.

C. Three descriptive parameters

The numerical values of the viewpoints that have positive and negative effects on productivity are used as explanatory parameters to be used when calculating the proposed productivity indicator. The three descriptive parameters are described below:

- Person-month payment rate (a): This is the basis of the cost paid to the contractor. The ratio of the person-month payment to that of a certain reference project can be used as the person-month payment rate.
- Feature-used rate (b): This is the rate of features actually used. It has been reported that 64% of features are not used in waterfall development [17], so the feature-used rate in this case is 0.36. Although 1.0 is ideal for agile development as we stack MVPs (Minimum Viable Products), the feature-used rate is expected to be less than 1.0 as features are culled when the software is finally released.
- NVA-NNVA rate (c): This is the rate of effort that does not contribute to value building, which can be described as the rate of wasted work in Lean. As shown in Table I, NVA is associated with the effort required to fix bugs that have been fed back from customers. In the case of waterfall development, 0 is acceptable because there is no feedback from customers during the development period. Verification and environment preparation are examples of NNVA.

These three descriptive parameters can be calculated regardless of whether development methodologies are waterfall or agile. In addition, it is possible to perform independent evaluations over time and to perform comparative evaluations on a project-by-project basis. In particular, the descriptive parameters a and c are factors that make agile development expected to be less productive than waterfall development as described in the introduction.

D. Proposed Productivity Indicators (delivery and cost indicators)

A delivery indicator $I_{delivery}$ and a cost indicator I_{cost} for evaluating productivity of software development are defined by Equations 1 and 2. These equations use three parameters

introduced in the previous subsection: person-month payment rate a, feature-used rate b, and NVA-NNVA rate c.

$$I_{delivery} = \frac{1}{(1-c) \times b} \quad (1)$$

$$I_{cost} = I_{delivery} \times a = \frac{a}{(1-c) \times b} \quad (2)$$

The term indicator $I_{delivery}$ described in Equation 1 defines the period during which value 1 is constructed after eliminating the probability of creating unused features and performing work that does not create value. The cost indicator I_{cost} described in Equation 2 is obtained by multiplying the delivery indicator and person-month payment rate. Since these values are calculated by multiplying three parameters that can be compared from the same viewpoint and side by side, any projects can be evaluated and compared. These two indicators are collectively called “productivity indicators.”

E. Proposed Productivity Ratio (delivery and cost ratio)

A delivery ratio $R_{Delivery}^k$ and a cost ratio R_{cost}^k are used for evaluation between multiple projects as shown by Equations 3 and 4.

$$R_{Delivery}^k = I_{delivery}^k / I_{delivery}^{ref} \quad (3)$$

$$R_{cost}^k = I_{cost}^k / I_{cost}^{ref} \quad (4)$$

Each ratio is a metric calculated by dividing the index calculated by Equations 1 and 2 by the value of a reference project. The ratio range from 0 to 1.0. The closer the value is to 0, the more productive it is than the reference project. In this case, less cost and less time can achieve more value.

IV. VERIFICATION EXPERIMENT

We evaluate the productivity of 20 agile trials using the proposed productivity indicators. We also evaluate the productivity of agile and waterfall developments using data of 20 agile trial projects and more than 200 waterfall development projects between 2014 and 2018.

A. Prerequisites

All 20 agile trial projects are delegated developments. Data at the end of December in 2019 were used for the assessment, so some projects were not completed. The sprint duration is 2-4 weeks. In each project, the effort of each work was linked to a work category based on the concept of Lean. In the past waterfall development projects, consideration, production, verification, and environment preparation were separately counted as the human-hours.

B. Productivity comparison among 20 agile trial projects

The descriptive parameters except feature-used rate b were measured for all 20 projects including those in the middle stage, and the delivery indicators and cost indicators were calculated. We provide all description parameters in Appendix. Selected

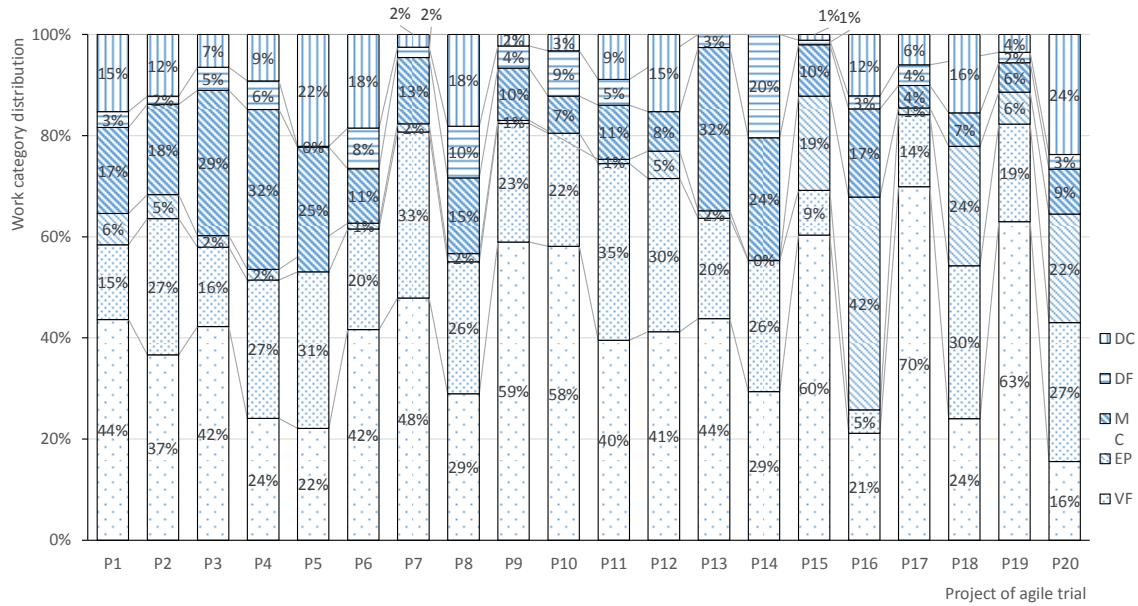


Fig.2 Work effort distribution in agile trials.

projects are shown in the figures 3-6. The feature-used rate b is set as 1.0 for convenience. Figure 2 shows the breakdown rate of each work category in the 20 agile trials. Comparing descriptive parameters of each project, Figures 3 and 4 show the person-month payment rate and NVA-NNVA rates, respectively, where the reference project is P8. The person-month payment rate ranges from 0.9 to 1.28, and the difference between the maximum and minimum is 0.38. On the other hand, when the NVA-NNVA rate is compared, the range is from 0.52 to 1.42, and the difference between the maximum and minimum values is 0.9. It can be seen that the difference is larger for work effort that does not contribute to value than for person-month payment. Figures 5 and 6 show delivery and cost ratios, respectively, where the reference project is P8. These four figures suggest:

- P11 has the highest person-month payment, while P5 and P10 have the lowest.
- P18 is the most wasteful, while P17 is the least.
- P18 takes the longest time to build the same value, while P17 takes the shortest.
- P18 is the most expensive project to build the same value, while P9 is the least.

Generally, we tend to evaluate productivity by person-month payment, but it is sometimes inaccurate from the viewpoint of Lean. Now, we pay attention to P14 and P17. Both projects have almost the same person-month payments, while both delivery and cost rates are significantly different. This is because P17 is much more efficient than P14 from the viewpoint of value constructing, so the difference in delivery and cost rates between two projects is as wide as 0.39 and 0.43, respectively.

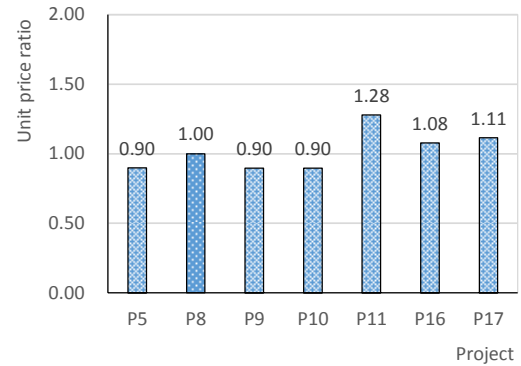


Fig.3 Person-month payment rate with reference to P8.

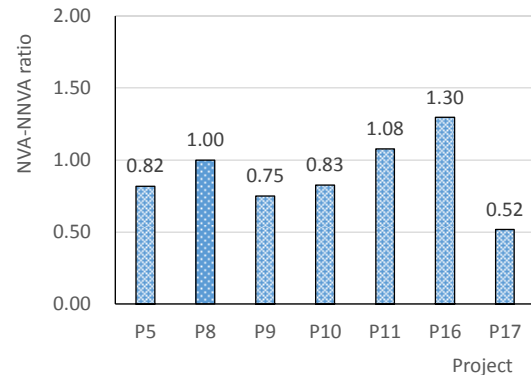


Fig.4 NVA-NNVA rate with reference to P8.

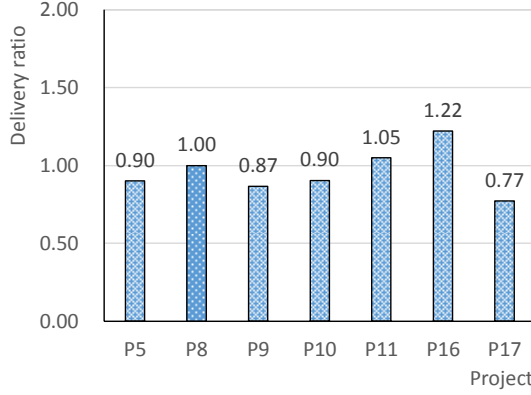


Fig. 5 Productivity comparison using “delivery rate” with reference to P8.

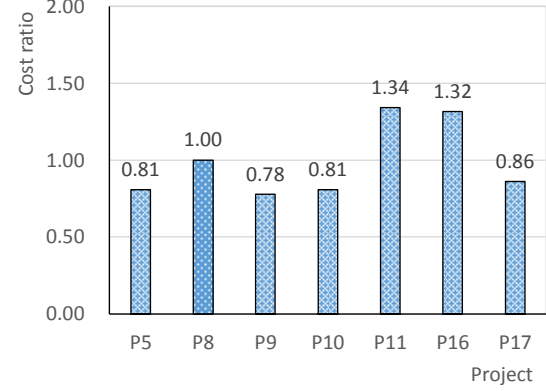


Fig. 6 Productivity comparison using “cost rate” with reference to P8.

In this way, the proposed productivity indicators show that multiple projects can be evaluated independently. By introducing downside of agile development, intuitive productivity indicators are provided.

C. Productivity Comparison with Past Waterfall Projects

We compared the productivity of 20 agile trials and more than 200 waterfall developments between 2014 and 2018 using statistics. Figure 7 indicates the average and median values for each work category in agile trials and waterfall developments. Table II shows the two descriptive parameters other than feature-used rate b for the agile trials and waterfall projects. The feature-used rate is set as 1.0 for convenience. Table III shows the delivery and cost rates.

TABLE II. DESCRIPTIVE PARAMETERS FROM STATISTICS

Project	Descriptive parameters	
	Person-month payment rate a	NVA-NNVA rate c
Agile trials	1.26	0.35
Waterfall	1.00	0.45

TABLE III. PRODUCTIVITY INDICATORS FROM STATISTICS

Project	Productivity indicators	
	$I_{delivery}$	I_{cost}
Agile trials	1.52	2.35
Waterfall	1.82	2.24

Tables II and III indicate the following:

- As expected, the person-month payment rate is 1.26 times higher for agile development than for waterfall development.
- The NVA-NNVA rate c was expected to show a higher validation rate in agile trials due to the higher validation frequency than waterfall development, but there was no significant difference between the two.
- Agile trials require 0.84 times the time and 1.05 times the outsourcing cost to achieve the same value as waterfall development.

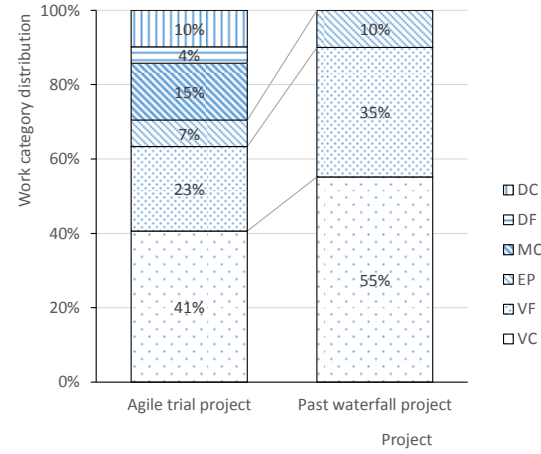


Fig. 7 Statistical work effort distribution of agile trials and past waterfall projects.

D. Productivity simulation using feature-used rate

Here, we have another simulation to clarify productivity of agile development compared with that of waterfall development by using the feature-used rate as a variable parameter. The advantage of agile development is to minimize waste by accumulating necessary functions, i.e., MVPs. Agile development is expected to have a higher feature-used rate than waterfall development.

The delivery and cost rates described in Equation 3 and 4 can be expressed as shown below:

$$\begin{aligned}
 R_{delivery}^{agl} &= I_{delivery}^{agl} / I_{delivery}^{wf} \\
 &= \frac{1 - c_{wf}}{1 - c_{agl}} \times \frac{b_{wf}}{b_{agl}} \\
 &\equiv f_{delivery} \left(\frac{b_{agl}}{b_{wf}} \right) \\
 R_{cost}^{agl} &= I_{cost}^{agl} / I_{cost}^{wf}
 \end{aligned} \tag{5}$$

$$\begin{aligned}
&= \frac{a_{agl}}{a_{wf}} \times \frac{1 - c_{wf}}{1 - c_{agl}} \times \frac{b_{wf}}{b_{agl}} \\
&\equiv f_{cost} \left(\frac{b_{agl}}{b_{wf}} \right)
\end{aligned} \quad (6)$$

Equations 5 and 6 indicate that both delivery and cost ratios can be described as a function of b_{agl}/b_{wf} , which is called the feature-used ratio between agile and waterfall developments. The simulation results are shown in Figure 8 with variable feature-used ratio b_{agl}/b_{wf} , from 1.0 to 2.7. The feature-used ratio is 1, when the feature-used rate is the same for both agile trials and past waterfall projects. The feature-used ratio is 2.7, when feature-used rates of agile and waterfall are 100 and 36% [17], respectively.

Figure 8 shows that the same value can be achieved in agile development in less than 84% of the time and for less than 105% of the cost of waterfall development, which is the same result as in Table III. Figure 8 also indicates that if the feature-used rate of agile development is more than 5% higher than that of waterfall development, higher productivity can be achieved in terms of development costs.

V. CONCLUSION AND FUTURE WORK

We proposed new productivity indicators focusing on the construction of value on the basis of the Lean concept, which can be evaluated by incorporating evaluation viewpoints such as person-month payment, work that adds no value, and unused functions. Our proposal is not only a solution to the situations in which the conventional productivity evaluation metrics become meaningless as cohesiveness is connected with productivity but also can be used to promote agile development by comparing it with past waterfall developments. Although we could compare comprehensive tendency of two development methodologies, the results do not indicate productivity of the same project members.

In the future, we think that the difficulty level of the project and its relationship with revenue need to be incorporated into the indicators. This is because projects with high difficulty and projects that contribute to profits should not be evaluated as unproductive, even if the value is realized through long-term development.

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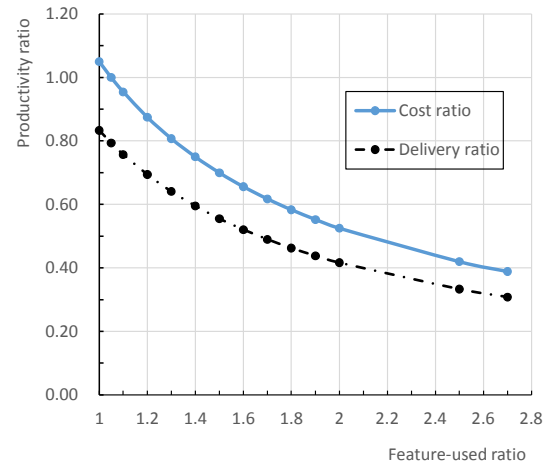


Fig.8. Comparison of agile trial and past waterfall projects at variable feature-used rates.

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APPENDIX : ALL DESCRIPTION PARAMETERS AND PRODUCTIVITY INDICATORS WITH REFERENCE TO P8

Project	Descriptive parameters		Productivity indicators	
	Person-month payment rate a *1	NVA-NNVA rate c *2	Delivery indicator	Cost indicator
P1	1.14	0.63	0.82	0.93
P2	1.14	0.88	0.93	1.06
P3	1.06	0.59	0.80	0.85
P4	1.05	0.93	0.96	1.00
P5	0.90	0.82	0.90	0.81
P6	1.02	0.77	0.88	0.89
P7	0.95	0.96	0.98	0.93
P8 (Reference)	1.00	1.00	1.00	1.00
P9	0.90	0.75	0.87	0.78
P10	0.90	0.83	0.90	0.81
P11	1.28	1.08	1.05	1.34
P12	0.95	0.94	0.97	0.91
P13	1.21	0.63	0.82	0.99
P14	1.12	1.22	1.16	1.29
P15	1.11	0.75	0.87	0.96
P16	1.08	1.30	1.22	1.32
P17	1.11	0.52	0.77	0.86
P18	1.07	1.42	1.34	1.44
P19	1.11	0.73	0.86	0.96
P20	1.07	1.37	1.29	1.38

*1 Person-month payment rate of all projects calculated for comparing the reference project P8.

*2 NVA-NNVA rate of all projects calculated for comparing the reference project P8.